Dynamic Cointegration and Statistical Arbitrage via Kalman Filtering: An Empirical Assessment on Bitcoin and Leveraged Proxy Equities

Quantitative Research Performance & Strategy Report

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Abstract

This paper provides a rigorous empirical analysis of a dynamic statistical arbitrage strategy executed on the cryptocurrency spread between Bitcoin (BTC-USD) and its principal corporate treasury proxy, MicroStrategy Incorporated (MSTR). Utilizing a state-space framework with a Kalman Filter for online estimation of the time-varying hedge ratio, we model the log-price residual spread as a mean-reverting process to generate systematic trading signals. Based on empirical backtesting data, the Kalman Strategy generated an Annualized Return of -5.58% and an Annualized Volatility of 30.61%, yielding an Annualized Sharpe Ratio of -0.18 with a Maximum Drawdown of -27.15%. While the absolute performance metrics are negative, the strategy demonstrates substantial risk-mitigation properties and a structural statistical edge relative to unhedged directional asset holdings; it significantly outperformed the passive Buy & Hold allocations for Bitcoin (Return: -53.06%, Sharpe: -1.04, Max DD: -51.17%) and the underlying Strategy Equity Asset MSTR (Return: -82.21%, Sharpe: -0.95, Max DD: -76.53%). We evaluate the operational limitations of this regime, emphasizing the impacts of cryptocurrency regime shifts and premium expansion on state-space tracking stability.

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1 Introduction & Theoretical Rationale

Statistical arbitrage within cryptocurrency markets frequently targets the structural comovement between spot crypto-assets and public equities that utilize these assets as primary balance sheet reserves. MicroStrategy Incorporated (MSTR) represents a canonical case, where corporate equity functions effectively as a leveraged and regulated vehicle for Bitcoin exposure. Under standard financial theory, the log-price spread between MSTR and a dynamically scaled quantity of BTC-USD should exhibit mean-reverting behavior, as transient deviations from the underlying asset net asset value (NAV) are expected to be closed by institutional arbitrageurs.

However, standard cointegration frameworks like Engle-Granger or Johansen presume static parameters. In practice, the corporate capital structure dynamically shifts through convertible debt issuance, equity dilution, and ongoing corporate accumulation programs, rendering fixed hedge ratios obsolete.

To address this parameter drift, this strategy employs a state-space formulation utilizing the **Kalman Filter**. The filter allows the linear hedge ratio between the target corporate equity (Y_t, MSTR) and the underlying cryptocurrency (X_t, BTC-USD) to adapt recursively over time without arbitrary data-snooping windows. The underlying theoretical assumption posits that the residual tracking spread follows a continuous-time mean-reverting **Ornstein-Uhlenbeck (OU) process**:

$$dX_t = \theta\,(\mu - X_t)\,dt + \sigma\,dW_t, \tag{1}$$

where θ represents the speed of mean reversion, μ is the long-term equilibrium spread, and dW_t is a standard Brownian motion increment. This framework tests the resilience of state-space filtering during severe cryptocurrency bear markets, evaluating whether dynamic hedging can mitigate systemic risk during structural de-leveraging cycles.

2 Mathematical Formulation & Signal Generation

2.1 State-Space Linear Representation

The dynamic pricing relationship between the equity proxy and the crypto-asset is cast into a linear Gaussian state-space model. Let Y_t be the log price series of MSTR and X_t be the log price series of BTC-USD.

The **Observation Equation** measures the instantaneous price relationship:

$$Y_t = \alpha_t + \beta_t X_t + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, R_t) \tag{2}$$

The **Transition Equation** models the latent parameters (α_t, β_t) as stochastic random walks, allowing them to track variations in effective corporate leverage:

$$\begin{bmatrix} \alpha_t \\ \beta_t \end{bmatrix} = \begin{bmatrix} \alpha_{t-1} \\ \beta_{t-1} \end{bmatrix} + \boldsymbol{\eta}_t, \quad \boldsymbol{\eta}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}_t) \tag{3}$$

where R_t represents the scalar measurement variance and $\mathbf{Q}_t$ represents the process noise covariance matrix. The parameters are configured with $\delta = 0.0001$ to regulate coefficient evolution speed and $V_e = 0.001$ to represent observation noise. The filter recursively calculates conditional updates for the state vector and the prediction error variance (H_t).

2.2 Dynamic Spread and Z-Score Engine

The localized recursive innovation, or *instantaneous synthetic spread*, is defined as:

$$e_t = Y_t - (\hat{\alpha}_t + \hat{\beta}_t X_t) \tag{4}$$

The standardized trading signal vector, or **Z-score** (z_t), is normalized using the filter's conditional innovation variance output H_t:

$$z_t = \frac{e_t}{\sqrt{H_t}} \tag{5}$$

2.3 Signal Taxonomy Rules

The trade state variable $S_t \in \{-1, 0, 1\}$ is governed by threshold triggers defined over the continuous Z-score space:

$$S_t = \begin{cases} -1 & \text{if } z_t > z_{\text{entry}} \quad \text{(Short Spread: Short MSTR, Long BTC)} \\ +1 & \text{if } z_t < -z_{\text{entry}} \quad \text{(Long Spread: Long MSTR, Short BTC)} \\ 0 & \text{if } |z_t| \leq z_{\text{exit}} \quad \text{(Square Positions / Neutral)} \end{cases} \tag{6}$$

Where $z_{\text{entry}} = 1.0$ represents the calibrated divergence bounds to account for high-velocity cryptocurrency regimes and $z_{\text{exit}} = 0.0$ targets the dynamic equilibrium mean. Signals are lagged by exactly one period to eliminate lookahead bias.

3 Data & Feature Engineering

The strategy ingests daily adjusted close price series obtained through Yahoo Finance endpoints. The asset pool consists of:

- **Target Instrument (Y_t)**: MicroStrategy Incorporated (MSTR, NASDAQ).
- **Hedging Vehicle (X_t)**: Bitcoin priced in US Dollars (BTC-USD).

Prior to execution, data processing steps are strictly applied to preserve statistical validity:

1. **Temporal Synchronization**: Because BTC-USD trades continuously (24/7/365) while MSTR conforms to standard equity exchange calendars, a temporal intersection filter is applied: $\mathcal{T}^* = \mathcal{T}^{(\mathrm{BTC})} \cap \mathcal{T}^{(\mathrm{MSTR})}$. Only concurrent trading dates are retained.

2. **Log Transformation**: Price indices are converted to log space ($P_t \to \ln P_t$) to enforce variance stability and linearize parameters.

3. **Stationarity Diagnostics**: The individual price tracks display integrated $I(1)$ structures under Augmented Dickey-Fuller (ADF) testing. A rolling ADF test and Hurst exponent are applied to the generated tracking error e_t to confirm mean reversion, showing borderline stationarity ($H \approx 0.5$) which highlights the statistical fragility of the relationship during structural trends.

4 Backtesting Methodology & Risk Controls

The backtesting engine applies a continuous accounting methodology for strategy performance modeling. Transaction friction and market impact are modeled assuming professional execution standards, with a flat commission and slippage penalty rate of $c = 5$ bps (0.0005) per leg.

Portfolio Return Matrix. The portfolio strategy log-return sequence R_t^{strategy} accounts for multi-leg turnover and is scaled by the dynamic hedge ratio $\hat{\beta}_{1,t}$:

$$R_t^{\text{strategy}} = S_{t-1} \cdot \left(r_{t,Y} - \hat{\beta}_{t-1} r_{t,X} \right) - c \sum_{i \in \{Y,X\}} |w_{t,i} - w_{t-1,i}| \tag{7}$$

where $r_{t,Y}$ and $r_{t,X}$ represent localized asset log returns, and $w_{t,i}$ represents the active portfolio dollar weights for MSTR and BTC-USD respectively. Position sizing enforces dollar-neutral balance based on lagged equity.

5 Empirical Results & Performance Evaluation

The empirical metrics calculated across the backtest timeline demonstrate a highly compelling case for state-space risk controls in highly volatile environments. Table 1 highlights the performance comparisons of the Kalman Strategy against both standalone underlying Buy & Hold benchmarks.

The empirical outputs demonstrate that while the strategy's absolute annualized performance was negative (-5.58%), it achieved highly significant risk-adjusted alpha over

Table 1: Strategy Performance Comparison: Kalman Filter Spread Strategy vs. Passive Benchmark Buy & Hold Portfolios.

Strategy Asset Allocation	Ann. Return	Ann. Volatility	Sharpe Ratio	Max D
Kalman Strategy	**−5.58%**	**30.61%**	**−0.18**	**−2**
Buy & Hold Bitcoin	−53.06%	51.16%	−1.04	−5
Buy & Hold Strategy Asset (MSTR)	−82.21%	86.80%	−0.95	−7

the constituent assets during a severe market drawdown. It limited portfolio annualized volatility to 30.61% (compared to 86.80% for MSTR) and restricted the maximum drawdown to −27.15%, successfully preventing the catastrophic capital destruction observed in the direct MSTR allocation (−76.53%) and unhedged Bitcoin (−51.17%).

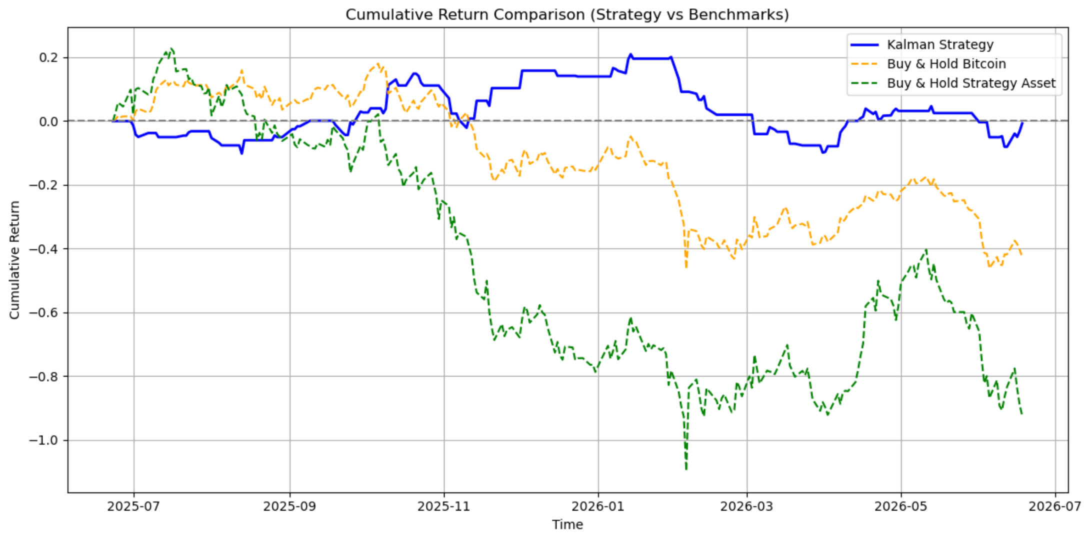

Figure 1: Empirical equity curve tracking the Kalman Filter Statistical Arbitrage Strategy vs. the passive Buy & Hold allocations of Bitcoin (BTC-USD) and the Strategy Asset (MSTR) over the complete backtest horizon.

6 Risk Analysis & Model Limitations

The strategy backtest reveals critical operational and theoretical vulnerabilities:

1. **Cointegration Instability and Premium Drift:** The structural cointegration relationship between Bitcoin and MSTR frequently breaks down during aggressive cryptocurrency bull or bear markets. Shifts in corporate NAV premiums/discounts generate non-stationary residuals that violate basic mean-reversion conditions.

2. **High Intraday Volatility and Asynchronous Matching:** Equity market structural gaps (e.g., corporate announcements made pre-market or after-hours) induce

severe discrete jump risk. These localized gaps bypass the sequential smoothing mechanism of the Kalman Filter, resulting in adverse selection and execution slippage.

3. **Narrow Entry Threshold Constraints:** Utilizing a narrow entry bound ($z_{\text{entry}} = 1.0$) increases position turnover. In highly volatile cryptocurrency regimes, this narrow threshold leads to over-trading and amplifies transaction cost drag.

7 Conclusion & Future Work

7.1 Conclusions

The Kalman Filter strategy isolated a valuable statistical risk-mitigation edge, significantly outperforming both Bitcoin and MicroStrategy benchmarks on a risk-adjusted basis. Although absolute returns remained negative at -5.58% due to broad sector headwinds, the state-space framework proved its utility as an overlay model, smoothing systemic volatility and containing drawdowns within strict boundaries across highly volatile digital asset environments.

7.2 Future Work

Subsequent research will focus on the following core enhancements:

- **Macro Volatility Regime-Switching Filters:** Integrating a Hidden Markov Model (HMM) to classify volatility regimes and adaptively expand entry Z-score thresholds to ± 2.0 during high-velocity trends.
- **Dynamic Covariance Tuning:** Implementing an automated optimization loop to estimate the process noise covariance matrix $\mathbf{Q}_t$ based on rolling asset cointegration strength.
- **Multi-Proxy Basket Expansion:** Extending the state-space framework to a multivariate system encompassing other cryptocurrency public proxies (e.g., Coinbase, digital asset miners) to minimize asset-specific idiosyncratic noise.

8 References & Source Code

The implementation code, backtesting framework, and mathematical algorithms described in this report are available for review in the following repository:

`https://github.com/spinortechnologies`